Choosing lahar volumes: the details



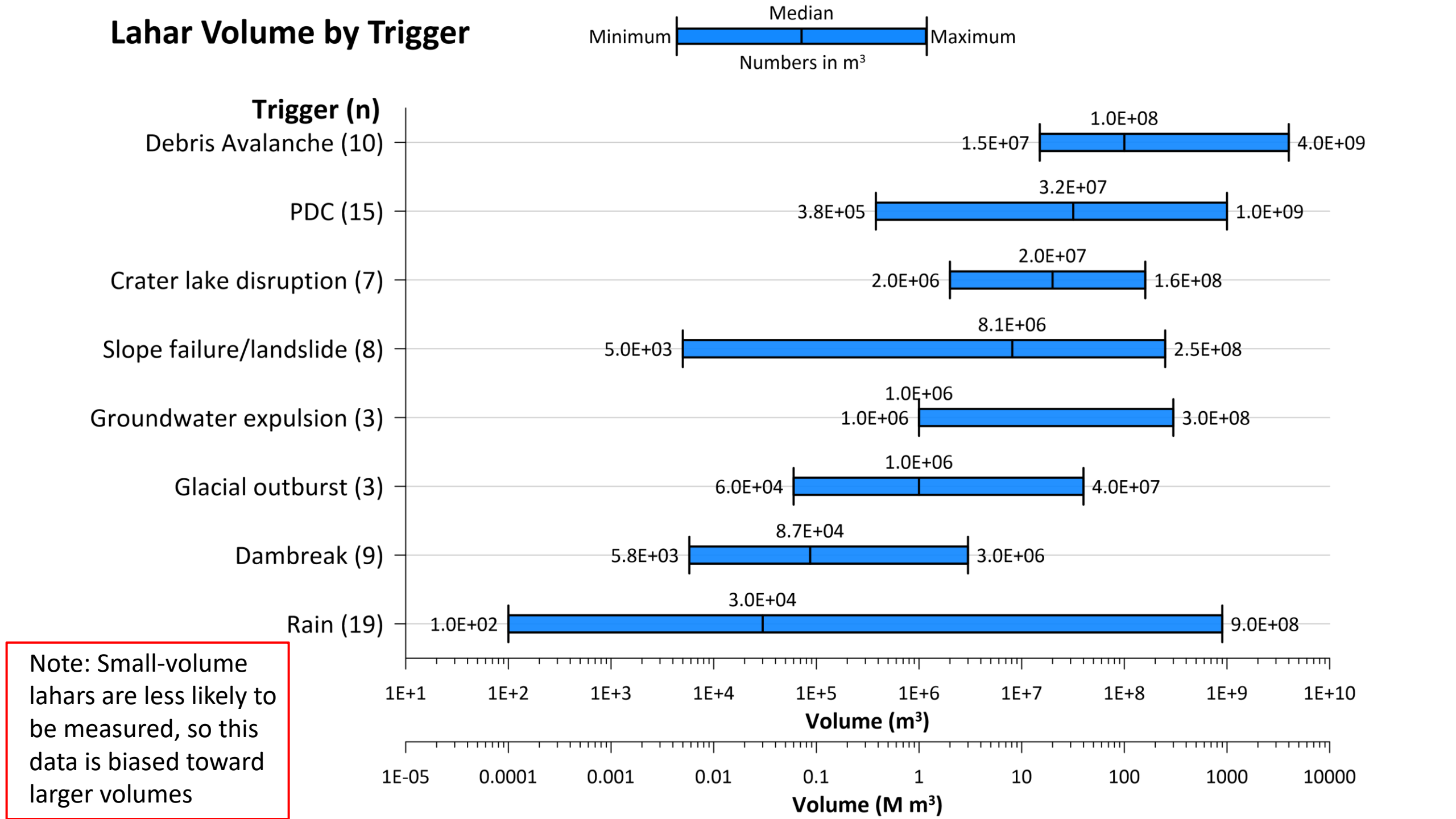
LAHARZ equations are based on very well measured lahars, most of which are $10^4 – 10^9$ m^3 (10,000 m^3 – 1 km^3) in volume

(Iverson et al. 1998)

TABLE 1. BASIC DATA RANKED IN ORDER OF FLOW VOLUME

Name of event	Location and date	Data source	Flow volume, V (m^3)	Inundated cross section Area, A (m^2)	Inundated planimetric area, B (m^2)
Osceola	Mount Rainier, United States,. 5000 B.P	Vallance and Scott, 1997	4.0×10^9	1.5×10^5	5.5×10^8
Teteltzingo	Citlaltepetl, Mexico, 18000 B.P.	Carrasco-Núñez et al., 1994	1.8×10^9	N.D.	1.4×10^8
Electron	Mount Rainier, United States, 530–550 B.P	Crandell, 1971	2.5×10^8	3.7×10^4	6×10^7
Round Pass	Mount Rainier, United States, 2700 B.P	Crandell, 1971	2×10^8	N.D.	5×10^7
Dead Man Flat	Mount Rainier, United States, 1100 B.P.	K. Scott et al., 1995; J.W. Vallance, unpublished data	1.8×10^8	1.4×10^4	9.0×10^7
National	Mount Rainier, United States, 500–1800 B.P.	K. Scott et al., 1995; J.W. Vallance, unpublished data	1.5×10^8	1.0×10^4	7.8×10^7
Paradise	Mount Rainier, United States, 4500–5000 B.P.	Crandell, 1971	1×10^8	N.D.	3.4×10^7
Zigzag	Mount Hood, United States, 1700 B.P	J. W. Vallance, unpublished data	7.3×10^7	1.2×10^4	5.5×10^7
Trout Lake	Mount Adams, United States, 6000 B.P.	Vallance, 1998	6.6×10^7	N.D.	2.7×10^7
Middle Fork Nooksack	Mount Baker, United States, 6000 B.P.	Hyde and Crandell, 1978	5×10^7	N.D.	2×10^7
Kautz Creek	Mount Rainier, United States, 1947	Crandell, 1971; Scott and Vallance, 1995	4×10^7	5000	4.5×10^6
Azufrado	Nevado del Ruiz, Columbia, 1985	Fritz et al., 1986; Pierson et al., 1990	4×10^7	2300	3.4×10^7
Molinos Nereidas (Chinchina)	Nevado del Ruiz, Columbia, 1985	Fritz et al., 1986; Pierson et al., 1990	3×10^7	1100	6.0×10^6
Guali	Nevado del Ruiz, Columbia, 1985	Fritz et al., 1986; Pierson et al., 1990	1.6×10^7	2000	1.1×10^7
Salt Creek	Mount Adams, United States, 200 B.P.	Vallance, 1998	1.5×10^7	N.D.	1.6×10^7
Tahoma	Mount Rainier, United States, 400–500 B.P.	Scott et al., 1995	1.5×10^7	1.9×10^4	6.0×10^6
Pine Creek + Muddy River	Mount St. Helens, United States, 1980	Pierson, 1985	1.4×10^7	2100	1.8×10^7
South Fork Toutle	Mount St. Helens, United States, 1980	Janda et al., 1981; Fairchild and Wigmosta, 1983	1.2×10^7	1500	3×10^7
Whitney Creek	Mount Shasta, United States, 1935	Osterkamp et al., 1986	4×10^6	N.D.	8×10^6
Bolum Creek	Mount Shasta, United States, 1897	Osterkamp et al., 1986	1.5×10^6	N.D.	3×10^6
Mabinit Eruption Lahars	Mayon, Philippines, 1984	Rodolfo, 1989	1.2×10^6	200	1.8×10^6
Tahoma Creek	Mount Rainier, United States, 1988	Walder and Driedger, 1994; J.S. Walder, unpublished data	6×10^5	190	1×10^6
Blue Lake	Mount St. Helens, United States, 1980	Major, 1984; Major and Voight, 1986	3.8×10^5	320	7.5×10^5
Butte Canyon	Mount St. Helens, United States, 1980	Major, 1984; Major and Voight, 1986	3.8×10^5	300	5.0×10^5
Mabinit Typhoon Saling	Mayon, Philippines, 1985	Rodolfo, 1989; Rodolfo et al.,1989	3×10^5	200	2×10^5
Middle	Mount St. Helens, United States, 1980	Major, 1984; Major and Voight, 1986	1.4×10^5	N.D.	4.0×10^5
Polallie Creek	Mount Hood, United States, 1980	Gallino and Pierson, 1984; T.C. Pierson, unpublished data	8×10^4	300	4.7×10^5
West Dodson	Columbia Gorge, Oregon, United States, 1996	R. M. Iverson, unpublished data	8×10^4	90	1×10^5
Mayflower Gulch	Tenmile Range, Colorado, United States, 1961	Curry, 1966	1.7×10^4	30	1.6×10^4

Lahar Volume by Trigger



Note: Small-volume lahars are less likely to be measured, so this data is biased toward larger volumes

Rainfall-triggered: Merapi


GVP

Merapi, Indonesia
After November 2010 eruption
<u>>1 million m³</u>

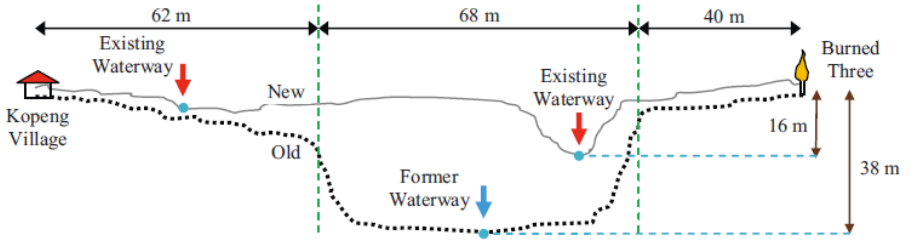


Figure 5: Rapid investigation measurement of a Kali Gendol cross section in Kopeng Village (±15 days after first eruption) showing the change in landscape caused by pyroclastic deposition.

Putra and Hariyadi, 2012

2010 PDCs filled 8 channels with unconsolidated material

Table 1: The eruption material of Mount Merapi which was concentrated in 8 surrounding Lahar Rivers (Mujiharjo *et al.* [1]).

No.	Lahar River	Pyroclastic deposit amount (m^3)
1	Boyong	8 million
2	Bebeng	10 million
3	Blongkeng	10 million
4	Woro	12 million
5	Kuning	14 million
6	Putih	18 million
7	Pabelan	24 million
8	Gendol	28 million
Total		124 million

Rainfall-triggered: Mayon

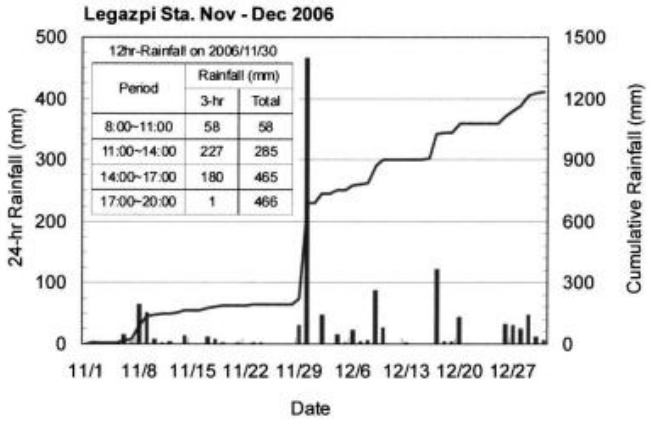


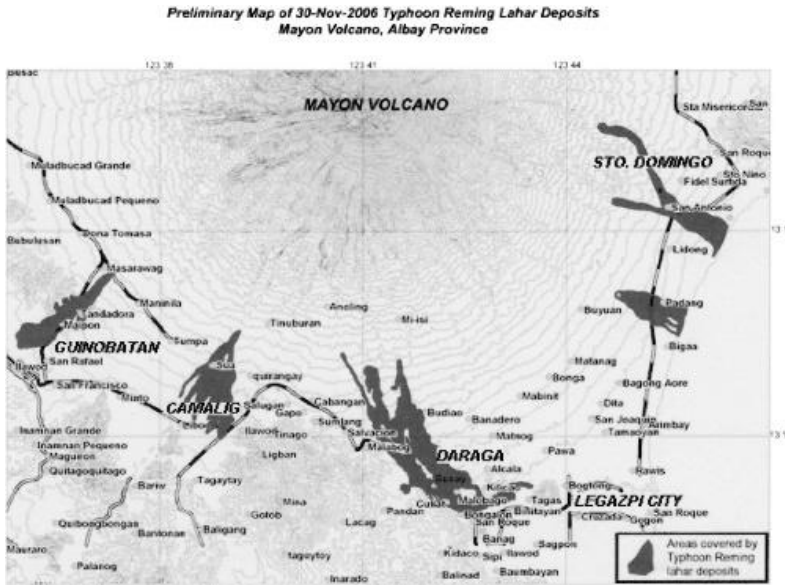
Fig. 5. Rainfall records for November-December 2006 in Legazpi City and rainfall details during the passage of Typhoon Reming (data from PAGASA)

Orense & Ikeda (2007)



Orense & Ikeda (2007)

Fig. 9. Scoured river bank in Barangay Travesia, Guinobatan Town. The dotted line on the photo indicates the former location of the river bank. The collapsed rail track can be seen at the background



Orense & Ikeda (2007)

Table 2 Calculated lahar volumes and deposit thicknesses exposed in gullies on the eastern and southern flanks of Mayon volcano

Volume calculation and deposit thickness

Area	Lahar deposit type	Max mapped thickness (m)	Min mapped thickness (m)	Volume (m^3)
Guinobatan–Camalig	Debris flow to hyperconcentrated flow	5.0	0.3	8,700,000
Daraga–Legaspi	Debris flow to hyperconcentrated flow	5.0	0.3	7,850,000
Padang	Debris flow to hyperconcentrated flow	1.6	0.1	850,000
Lidong	Debris flow to hyperconcentrated flow	1.5	0.2	300,000
San Antonio	Debris flow to hyperconcentrated flow	1.7	0.2	1,100,000
Total				18,800,000

Paguican et al. (2009)

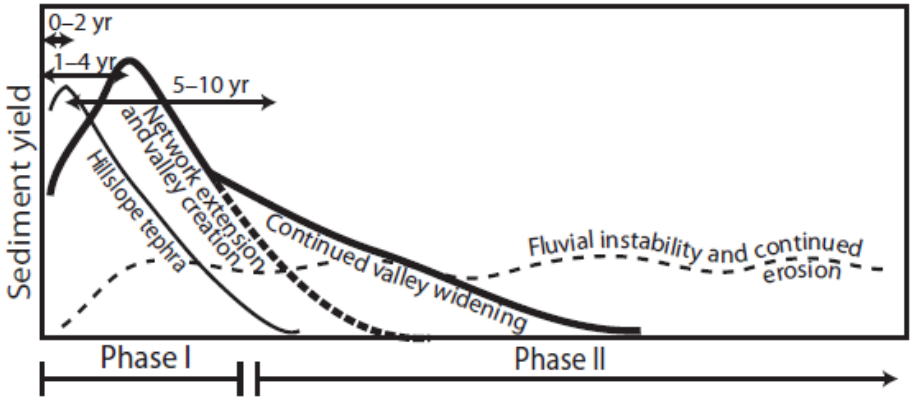
Mayon, Philippines
2006 (typhoon Reming)
<u>300,000 m³ to 9 million m³</u>
655 deaths, 2,437 injured and 445 missing

Rainfall-triggered: Pinatubo



A lahar, or volcanic mudflow, fills the banks of the Pasig-Potrero River on the east side of Pinatubo volcano in the Philippines on 13 October 1991. The lahar moved at a velocity of 3-5 m/s and carried a few meter-sized boulders. This lahar was not directly produced by an eruption, but was triggered by minor rainfall that remobilized thick deposits of ash and pumice that blanketed the landscape. Devastating lahars occurred at Pinatubo for years after the catastrophic 1991 eruption.

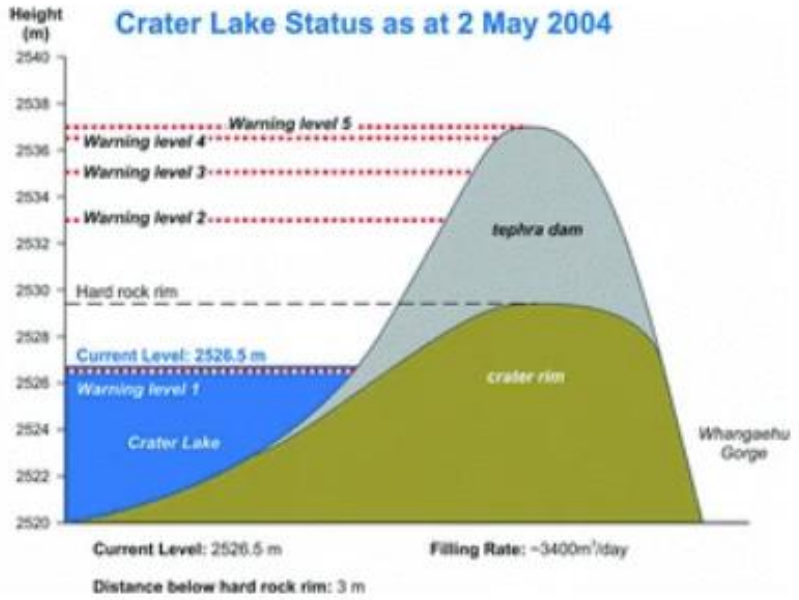
Photo by Chris Newhall, 1991 (U.S. Geological Survey).



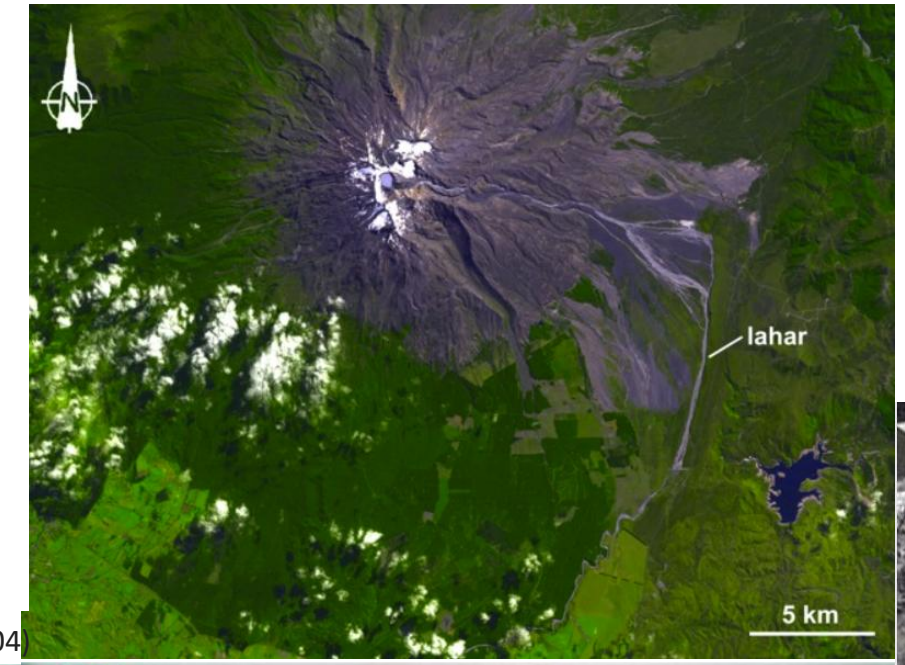


Pinatubo, Philippines
After 1991 eruption
1991: >900 million m³ (0.9 km³)
1992: >600 million m³ (0.6 km³)

Crater lake/dambreak



Wanganui-Rangitikei Emergency Management Committee (2004)



Ruapehu, New Zealand
March 18, 2007
4.4 million m³

GNS Science and Vern Manville





GNS Science and Vern Manville





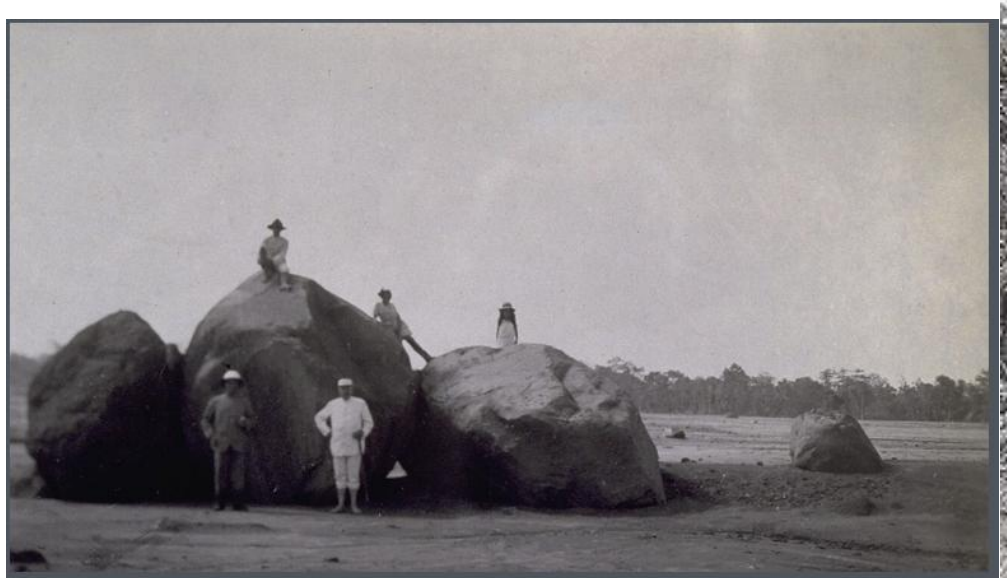
3:45-5:45

http://www.youtube.com/watch?v=5x5tZAHEoRU



Crater lake breach (PDCs)

Kelut, Indonesia
May 19, 1919
~**40 million m³**
5,110 deaths



Three months after the eruption, large boulders at Kemloko carried by the devastating lahars of the May 19-20, 1919, eruption capture the attention of visitors.

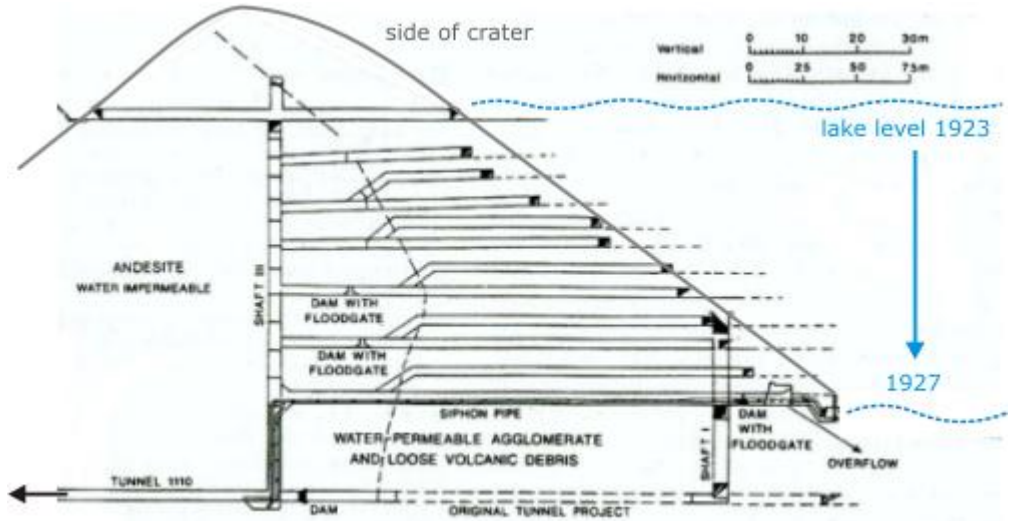
Photo from the collection of Maurice and Katia Krafft.

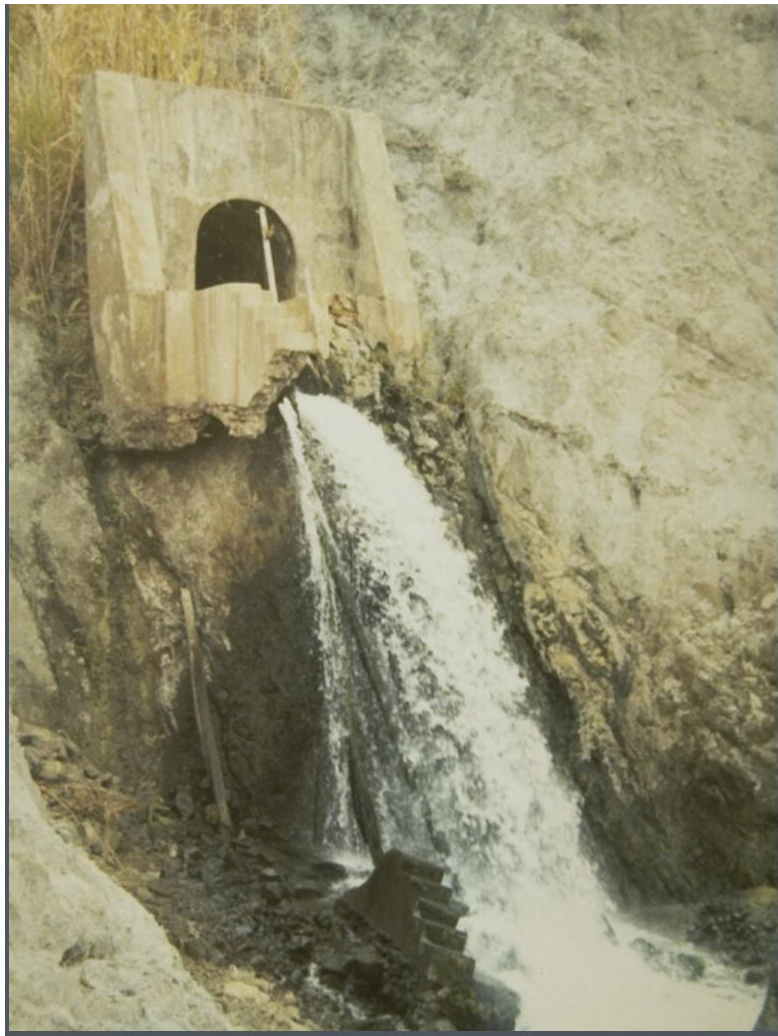




This August 1922 photo from the NW shows the still vegetation-free summit of Kelud volcano following the powerful May 1919 eruption and the relatively mild December 1920 eruption. The 1919 eruption produced devastating pyroclastic flows and lahars that killed over 5,110 people. This prompted an engineering project to construct tunnels to lower the level of the crater lake, which was completed in 1926. A milder explosive eruption in 1920 was accompanied by extrusion of a small lava dome beneath the crater lake.

Photo published in Taverne, 1926 "Vulkaanstudien op Java," (courtesy of Volcanological Survey of Indonesia).





The outlet tunnel of the Kelud crater lake drainage system in 1973. A series of drainage tunnels and shafts were constructed following the devastating 1919 eruption that killed 5,110 people to decrease the amount of water in the summit crater lake. Loss of life from devastating lahars produced by the explosive ejection of crater lake water has been significantly reduced in subsequent eruptions. The original outlet was lowered to this level following the deepening of the crater by an eruption in 1951.

Photo by Sumarma Hamidi, 1973 (Volcanological Survey of Indonesia).

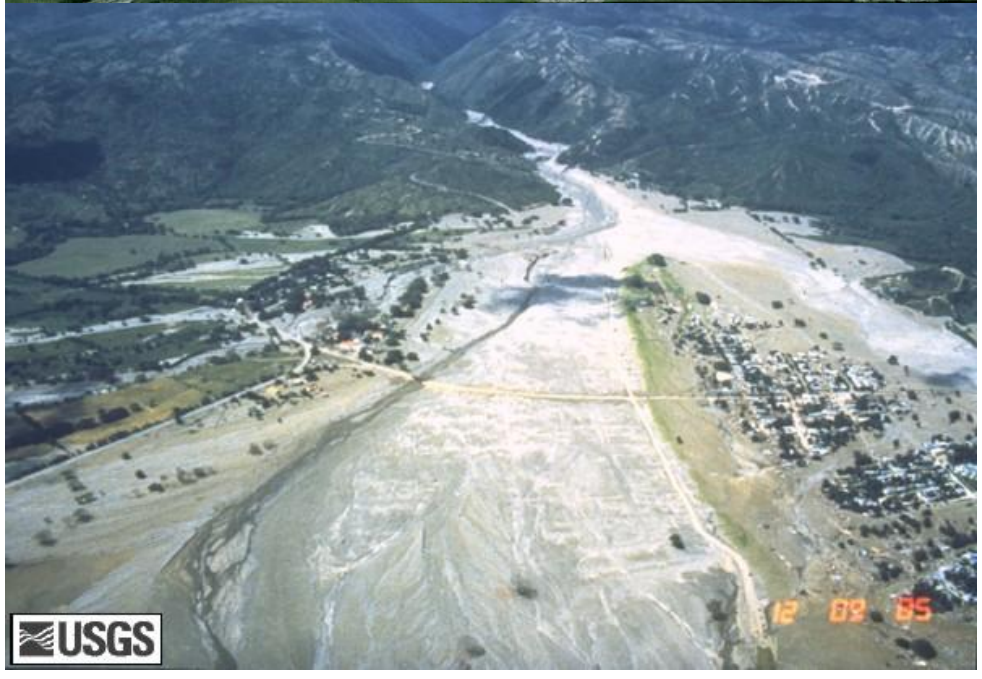
PDC-snow/ice melt

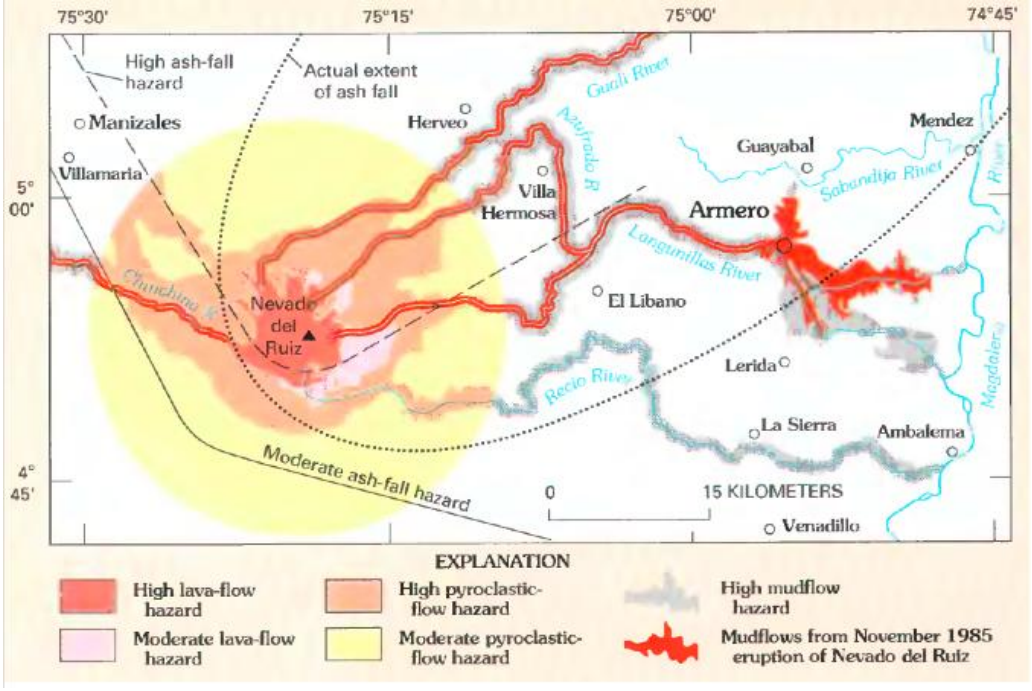
Nevado del Ruiz, Colombia
November 13, 1985
40 million m³ (Azufrado)
30 million m³ (Molinos Nereidas- Chinchina)
16 million m³ (Guali)
>23,000 deaths, >5000 injuries





VDAP





PDC-glacier melt





This massive boulder on the SW side of Cotopaxi was carried in a lahar, possibly during a major eruption in 1877. Scale is provided by volcanologists Minard Hall and Patty Mothes. More than 130,000 people live in areas subject to lahar risk from Cotopaxi. The 1877 eruption produced lahars that covered this valley, swept into eastern river drainages, and reached the Pacific Ocean along valleys to the NW.

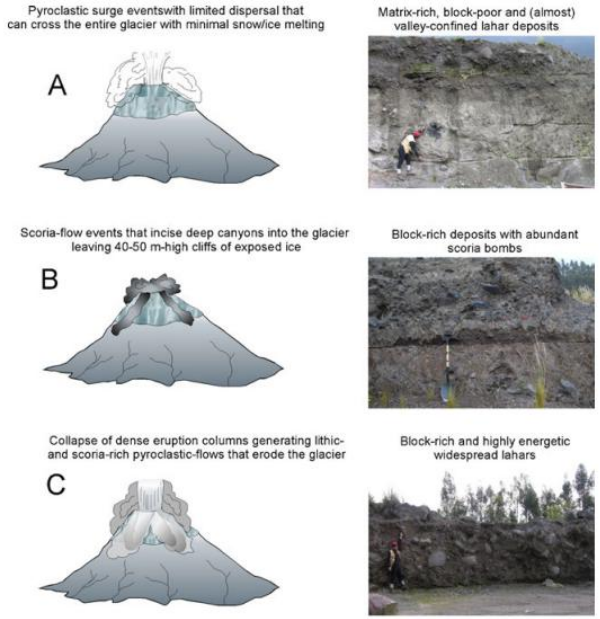
Photo by Tom Pierson, 1992 (U.S. Geological Survey).

Cotopaxi, Ecuador
June 26, 1877
70-90 million m³

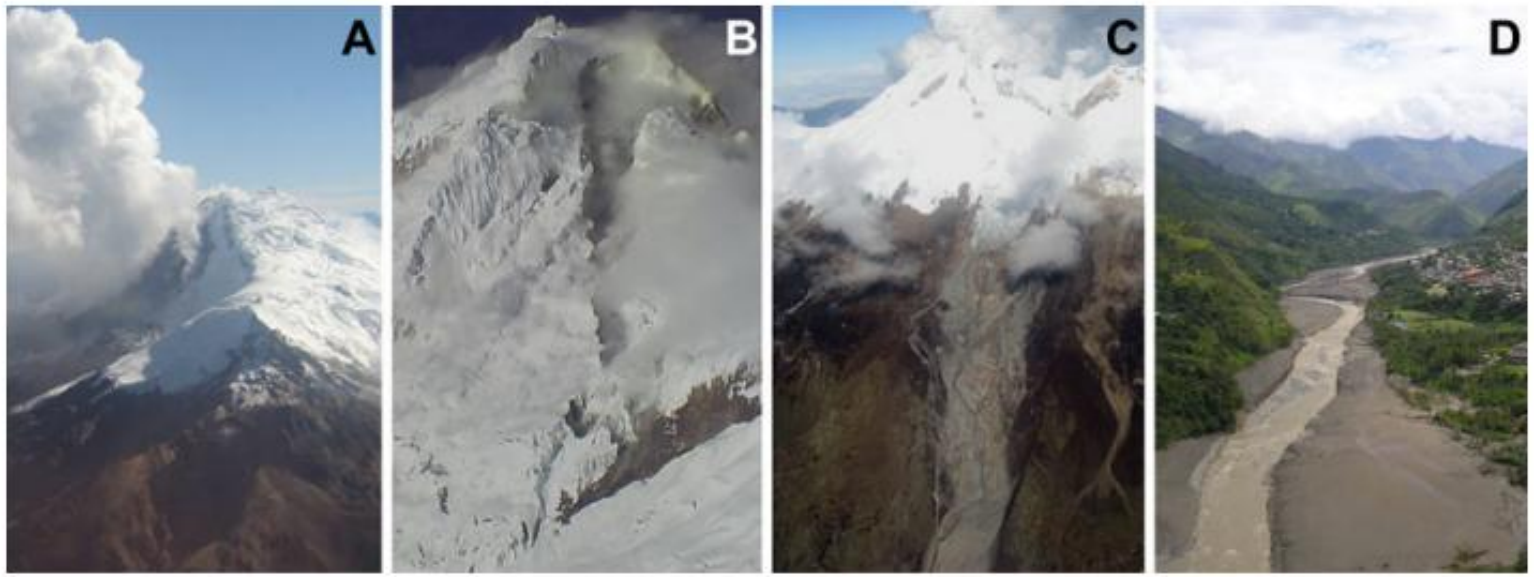


Fig. 11 General sketch of different types of interaction between the explosive event and the glacier and corresponding lahar deposits. **a** Dilute pyroclastic–density–currents that generate shallow ice melting and matrix-rich, block-poor lahar deposits. **b** Deep, but overall confined ice erosion (e.g., 1877 event). **c** Deep and widespread ice erosion

Pistolesi et al. (2013)



Groundwater expulsion





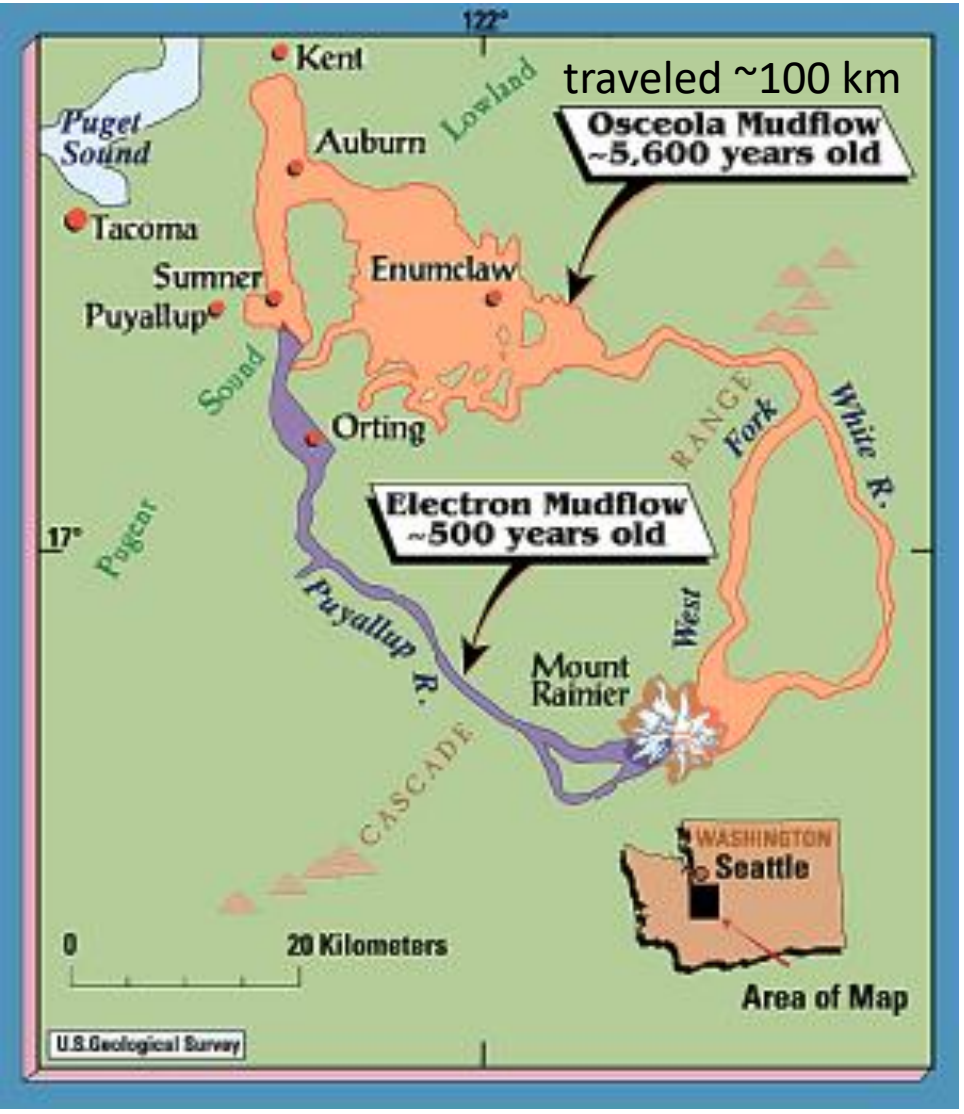
Travelled ~150 km

Fig. 2 **a**, **b** Fissures that opened after the February and April 2007 eruptions. **c** Traces of the destructive lahar in April 2007. View of the east flank of Nevado del Huila where the lahar flowed into the Simbola River. **d** View upstream of Belalcázar (*right side*) and beyond. Belalcázar is the biggest town in the Páez River valley that was hit by the flow. (Photographs by INGEOMINAS)

Worni et al. (2011)

Nevado del Huila, Colombia
February 18, 2007
April 18, 2007
November 21, 2008
>375 million m³

Debris avalanche-triggered



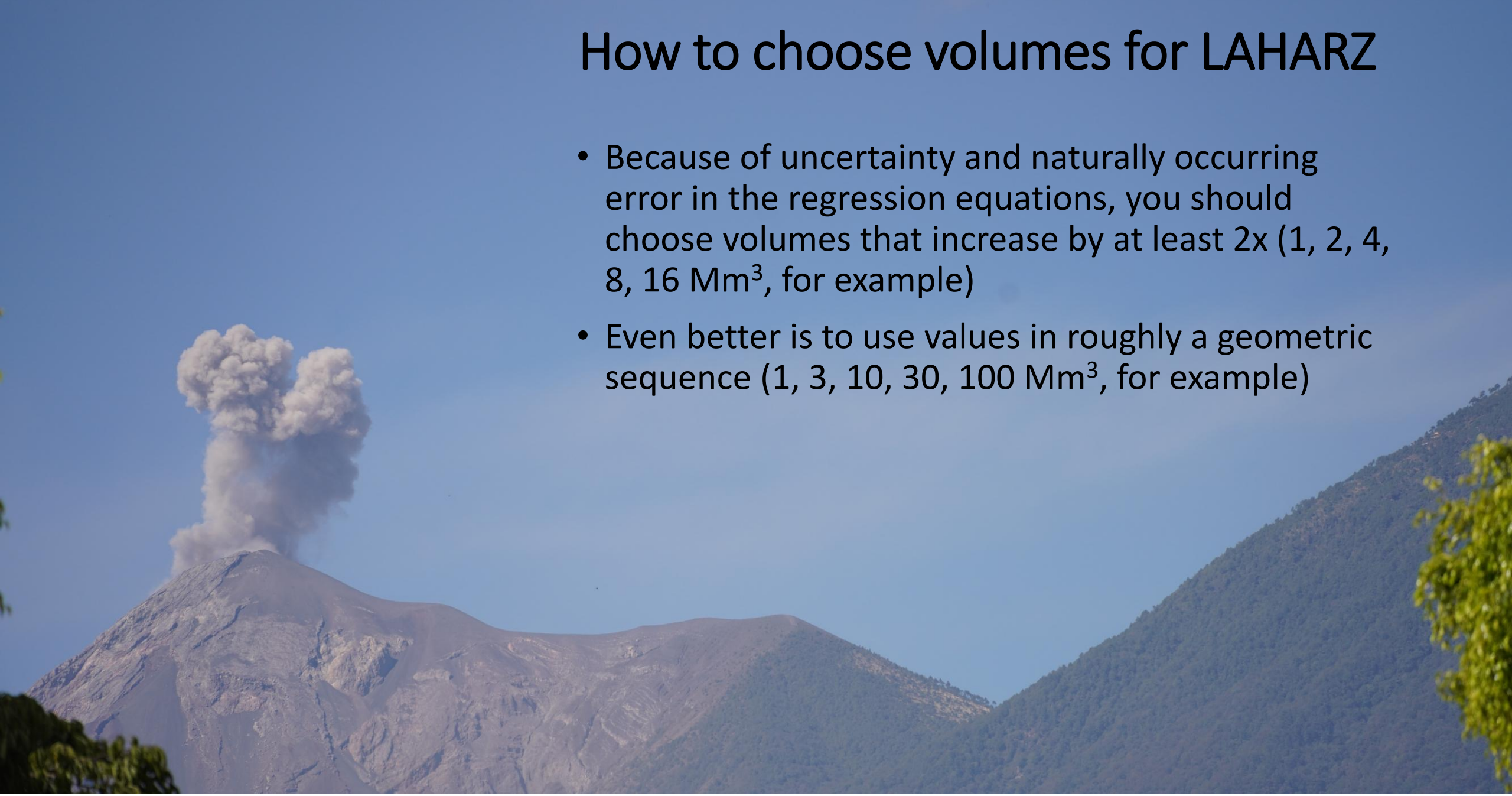


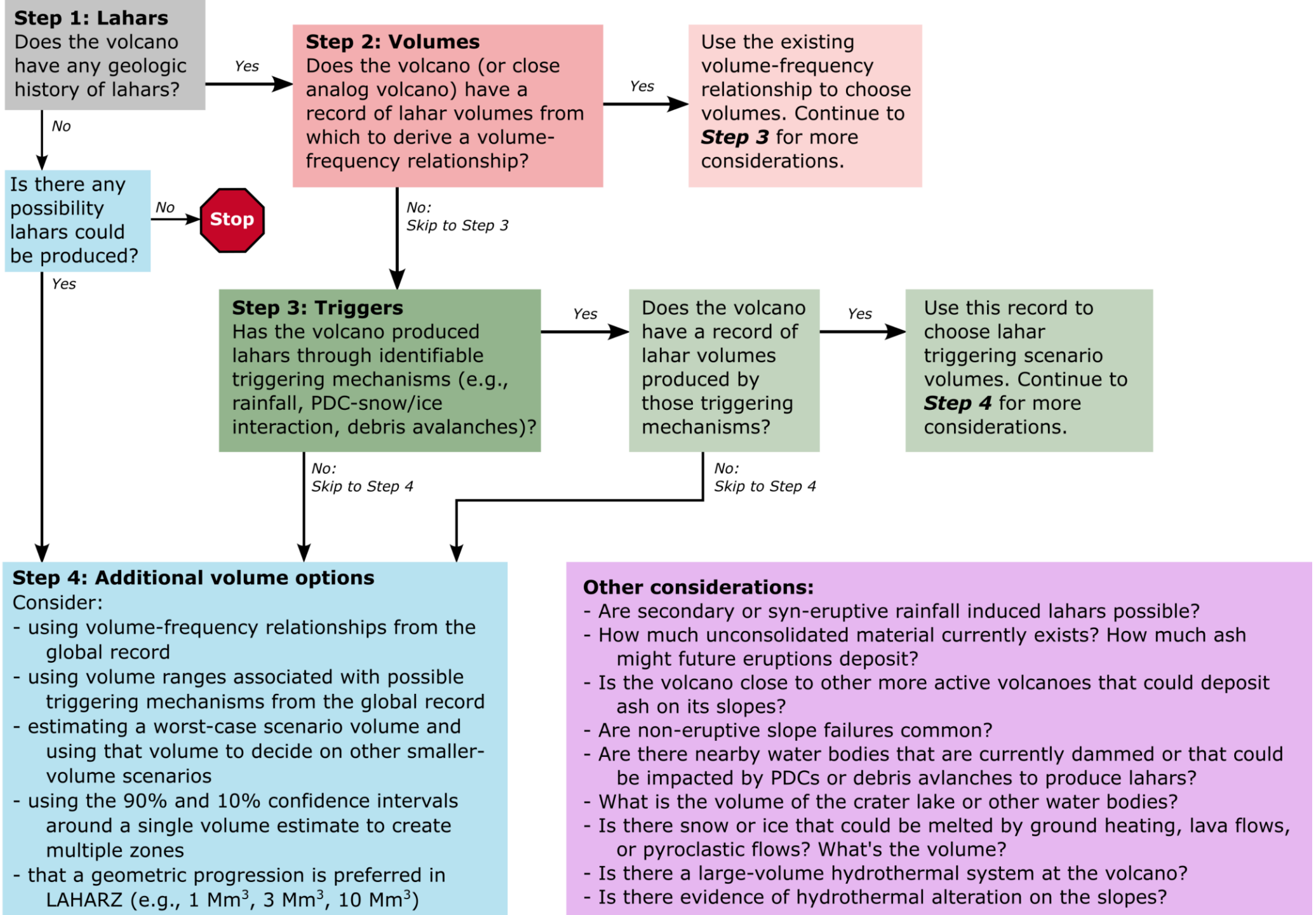
Mount Rainier, USA
~5600 yr ago
<u>>4000 million m^3 (4 km^3)</u>



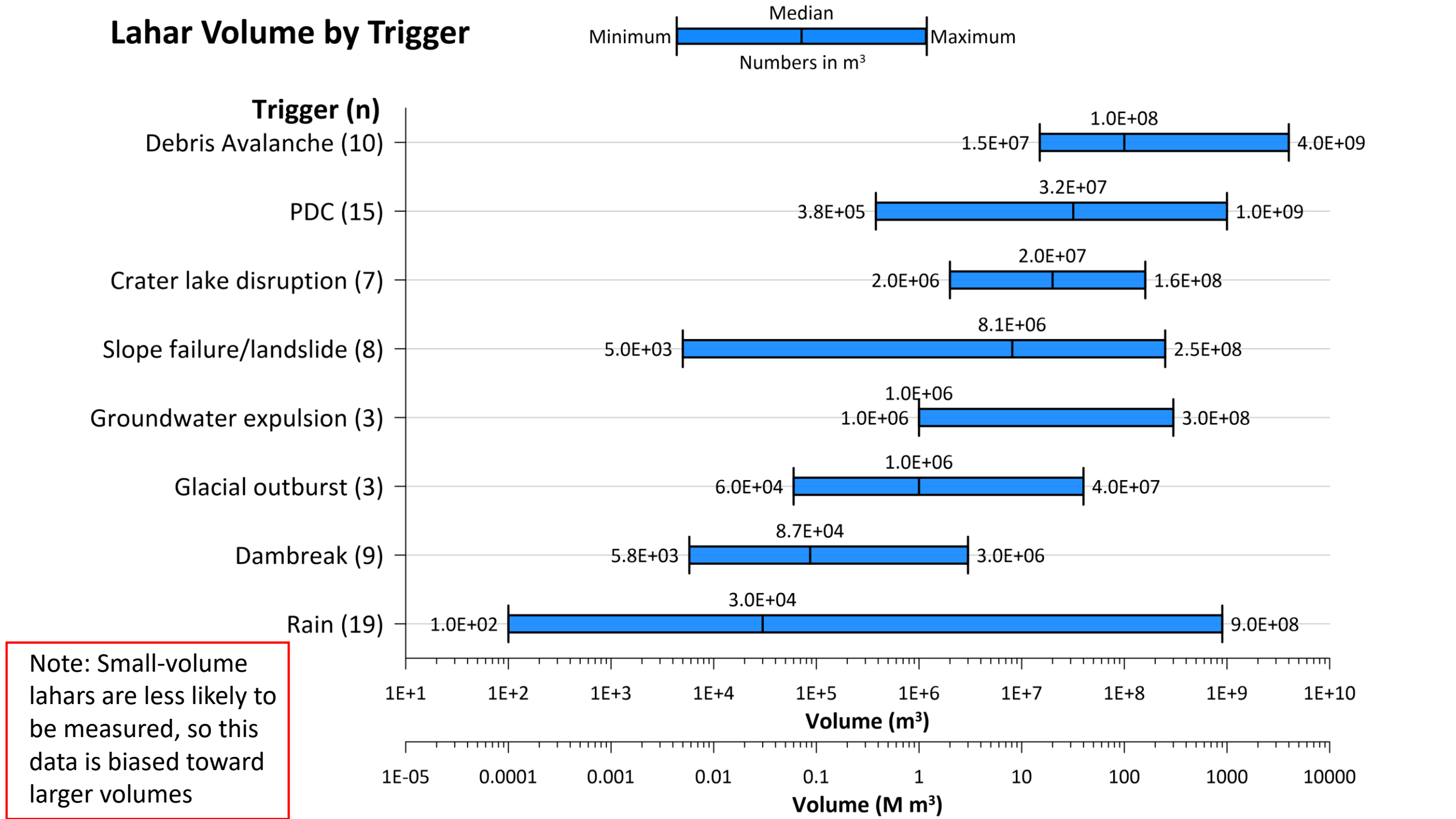
How to choose volumes for LAHARZ

- Because of uncertainty and naturally occurring error in the regression equations, you should choose volumes that increase by at least 2x (1, 2, 4, 8, 16 Mm^3, for example)
- Even better is to use values in roughly a geometric sequence (1, 3, 10, 30, 100 Mm^3, for example)




Step 1: Lahars
Does the volcano have any geologic history of lahars?
Yes
No
Step 2: Volumes
Does the volcano (or close analog volcano) have a record of lahar volumes from which to derive a volume-frequency relationship?
Yes
Use the existing volume-frequency relationship to choose volumes. Continue to Step 3 for more considerations.
No:
Skip to Step 3
Is there any possibility lahars could be produced?
No
Stop
Yes
Step 3: Triggers
Has the volcano produced lahars through identifiable triggering mechanisms (e.g., rainfall, PDC-snow/ice interaction, debris avalanches)?
Yes
Does the volcano have a record of lahar volumes produced by those triggering mechanisms?
Yes
Use this record to choose lahar triggering scenario volumes. Continue to Step 4 for more considerations.
No:
Skip to Step 4
No:
Skip to Step 4
Step 4: Additional volume options
Consider:
- using volume-frequency relationships from the global record
- using volume ranges associated with possible triggering mechanisms from the global record
- estimating a worst-case scenario volume and using that volume to decide on other smaller-volume scenarios
- using the 90% and 10% confidence intervals around a single volume estimate to create multiple zones
- that a geometric progression is preferred in LAHARZ (e.g., 1 Mm3, 3 Mm3, 10 Mm3)
Other considerations:
- Are secondary or syn-eruptive rainfall induced lahars possible?
- How much unconsolidated material currently exists? How much ash might future eruptions deposit?
- Is the volcano close to other more active volcanoes that could deposit ash on its slopes?
- Are non-eruptive slope failures common?
- Are there nearby water bodies that are currently dammed or that could be impacted by PDCs or debris avlanches to produce lahars?
- What is the volume of the crater lake or other water bodies?
- Is there snow or ice that could be melted by ground heating, lava flows, or pyroclastic flows? What's the volume?
- Is there a large-volume hydrothermal system at the volcano?
- Is there evidence of hydrothermal alteration on the slopes?

Lahar Volume by Trigger



Note: Small-volume lahars are less likely to be measured, so this data is biased toward larger volumes

How to choose volumes for LAHARZ

- Check historical records of lahar volumes at your volcano
  - What types of lahars are produced at your volcano?
    - Rainfall-triggered?
    - Eruption-triggered (Snow melt? Ice melt? Groundwater expulsion? Damming of rivers by deposits? Flows impacting lakes or rivers? Crater lake expulsion?)
  - What is the most common lahar volume?
  - What is the worst-case scenario (biggest) lahar volume?
- Investigate the amount of snow and ice on your volcano
- Investigate whether any crater lakes are present and what volume they are
- Investigate the rainfall/typhoon/hurricane history at your volcano
  - Is there a rainy season?
- Check historical records at analog volcanoes
  - Volcanoes of a similar size
  - Volcanoes with similar amounts of snow/ice
  - Volcanoes with similar eruptive histories (VEI and PDCs)
  - Volcanoes with similar crater lakes
  - Volcanoes in similar climates